IBM SkillsBuild – AICTE-2026

PROJECT PROBLEM STATEMENT

Problem Statement ID: 17

Problem Statement Title: Multi-language Course Content Translator Agent

Domain: Education Technology (EdTech) · Generative AI / Natural Language Processing

Problem Statement:

Students often receive technical course material in a language that is not their strongest, which slows learning. Copying such content into a general translator breaks its structure: code and formulas get altered, numbers and technical terms change, and headings and lists are lost. The problem is to build a Multi-language Course Content Translator Agent that accepts course documents (PDF, DOCX, PPTX, TXT), translates the content between English, Hindi, Marathi and other languages using IBM Granite, preserves technical terms, mathematical expressions, numbers, code, headings and lists as accurately as possible, and helps students understand difficult concepts in simple language. The project addresses this problem.

Project Name: Bhasha Bridge

Objectives / Requirements:

- Upload PDF, DOCX, TXT and PPTX study documents, with safe file validation and size limits.
- Extract and display the document text correctly.
- Translate academic content between English, Hindi, Marathi and other supported languages.
- Preserve technical terms, mathematical expressions, numbers, code, headings and lists.
- Display the original and translated content side by side.
- Provide a student-friendly simplification mode that explains difficult concepts in simple language.
- Provide a glossary of important technical terms.
- Allow translation of selected sections and download of the translated content.
- Include document history, clear error messages, loading states and sample documents.
- Use IBM Granite through IBM watsonx.ai (IBM Cloud Lite), with no hardcoded API keys, and explain translation uncertainty with a recommendation for human review.

Expected Outcome:

A working, responsive web application (FastAPI backend, HTML/CSS/JavaScript frontend) that lets a student upload a course document, translate it with IBM Granite into the language of their choice, compare original and translation side by side, get a plain-language simplification and a glossary, and download the result. The deliverables are the complete source code, setup instructions and .env.example, API documentation and tests, an architecture and data-flow description, an explanation of the IBM Granite and IBM Cloud Lite integration, a Responsible AI section, and a 2–3 minute demonstration script with a presentation outline.

Source:

IBM SkillsBuild – AICTE-2026 problem statements (Problem Statement #17)